E-Commerce App Design Project Report

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1. Introduction

This project presents the design of an interactive e-commerce mobile application prototype created in Figma. The goal was to apply core Human–Computer Interaction (HCI) principles—including user-centered design, usability, accessibility, and iterative refinement—to develop an intuitive digital shopping experience. The prototype demonstrates clear interaction flows, visual hierarchy, and efficient navigation consistent with user expectations for mobile commerce platforms.

2. Design Process

The design followed a user-centered, iterative process aligned with HCI best practices:

2.1 Requirement Gathering

Initial requirements were derived from typical user needs when shopping online:

- Browse items across multiple categories
- Quickly search for specific products
- View product details and images
- Add items to a cart and complete checkout smoothly
- Review order history and manage account information

2.2 Ideation & Sketching

Early sketches focused on:

- A clean home screen layout

- Category-based browsing
- A consistent product card design
- A simplified checkout flow

Sketching allowed rapid exploration of layout structures before committing to visual design.

2.3 Wireframing

Low-fidelity wireframes mapped out the essential screens:

- Home
- Category listings
- Product detail pages
- Cart
- Checkout
- Login and account pages

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These wireframes established the information hierarchy and ensured clarity before moving to high-fidelity design.

2.4 High-Fidelity Design

The visual design stage added:

- Color palette
- Typography
- Icons
- Polished UI components

- Image placeholders for products

The goal was to maintain clarity and aesthetic consistency while improving engagement and usability.

2.5 Prototyping

Interactive flows were built in Figma, enabling:

- Tap navigation
- Scrolling interactions
- Button and link transitions

2.6 User Testing & Refinement

A small group of users tested the prototype, completing browsing and checkout tasks. Their feedback guided refinements in navigation clarity, filter discoverability, and checkout smoothness.

3. User Research

3.1 Research Participants

Participants included 5 individuals aged 18–45 who shop online regularly.

3.2 Key Findings

Users expressed needs related to:

- Fast product discovery
- Clear product details and imagery
- Trustworthy, simple checkout steps
- Familiar navigation patterns
- Mobile-friendly gestures (scrolling, swiping images)

3.3 Insights from Research

- Users rely heavily on product images when deciding.
- Category organization must be clear and consistent.
- Filters and sorting tools are essential for efficiency.
- Overly complex forms reduce checkout completion.

4. Personas

These personas refer to accessing ecommerce websites.

Persona 1 – "The Quick Shopper"

Goals: Buy items quickly with minimal steps
Pain Points: Long checkouts, cluttered interfaces
Needs: Fast search, streamlined checkout

Persona 2 – "The Comparison Shopper"

Goals: Compare features and pricing
Pain Points: Limited information, unclear images
Needs: Detailed product pages, robust filtering

Persona 3 – "The Casual Browser"

Goals: Explore categories without a specific plan
Pain Points: Overwhelming layouts
Needs: Clean design, recommended items, simple navigation

5. Task Flows

The following major task flows were designed and prototyped:

5.1 Browsing Flow

Home → Category → Product → Add to Cart

5.2 Search Flow

Search → Filter/Sort → Product → Add to Cart

5.3 Checkout Flow

Cart → Shipping Details → Payment → Confirmation

5.4 Account Management

Login → Profile → Order History

Each flow emphasizes minimal steps, familiarity, and clear visual cues.

6. Wireframes

Low-fidelity wireframes established structure:

6.1 Home Screen Wireframe

- Banner/featured items
- Category menu
- Recommended product section

6.2 Category Page Wireframe

- Grid layout for product cards
- Filters (top right)
- Sort option

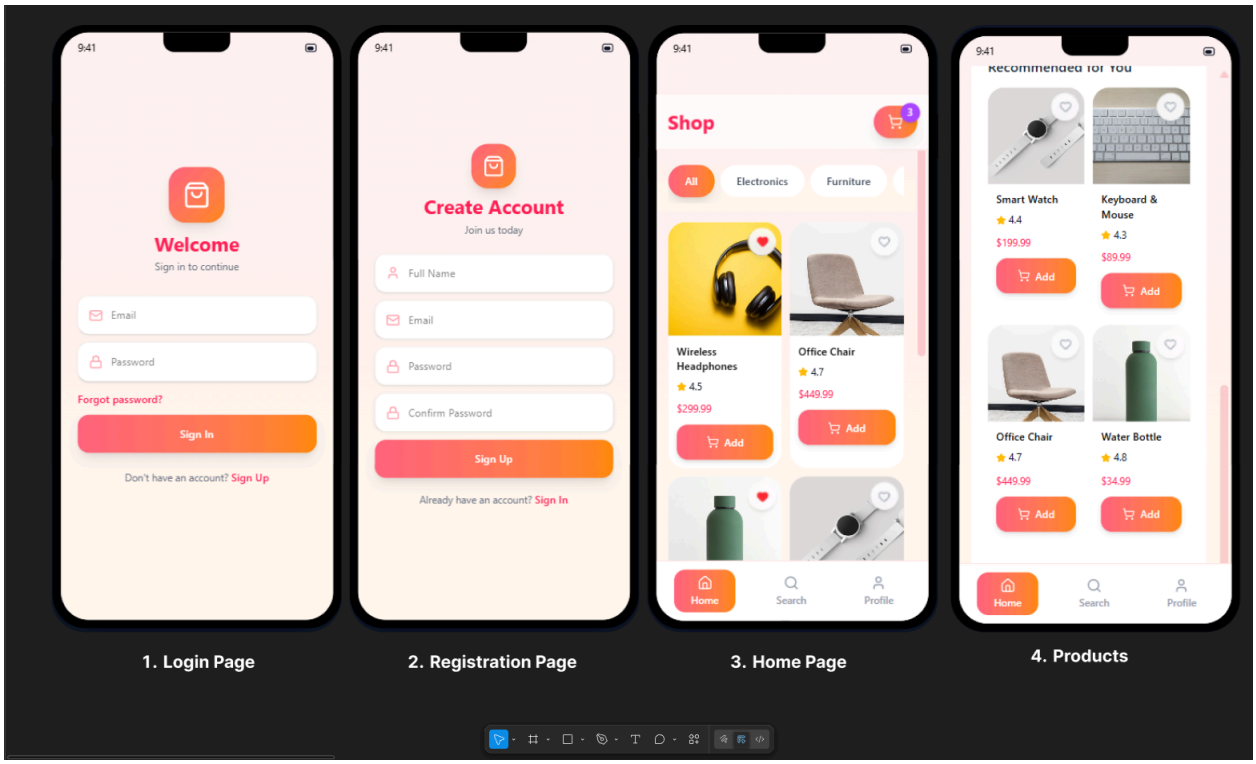
6.3 Product Detail Page Wireframe

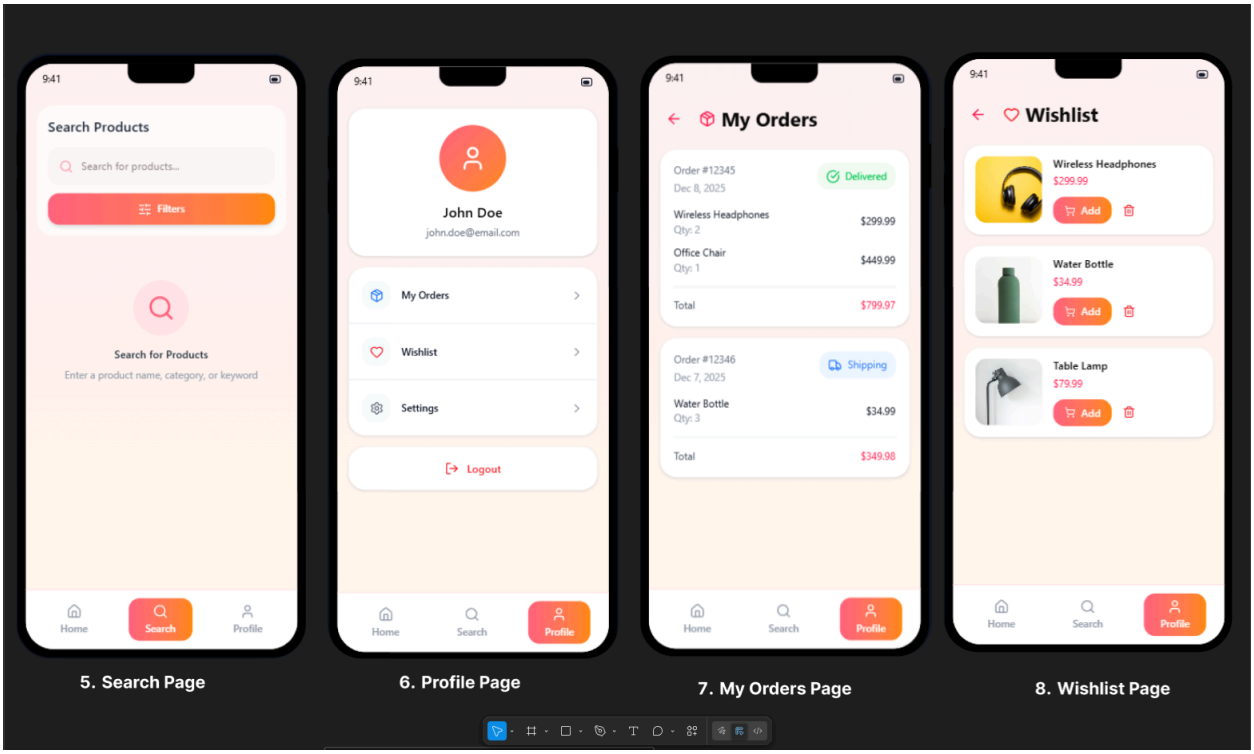
- Large primary image
- Additional image thumbnails

- Product description
- Add to Cart button

6.4 Cart & Checkout Wireframes

- Item list with quantities
- Price summary
- Simple checkout steps



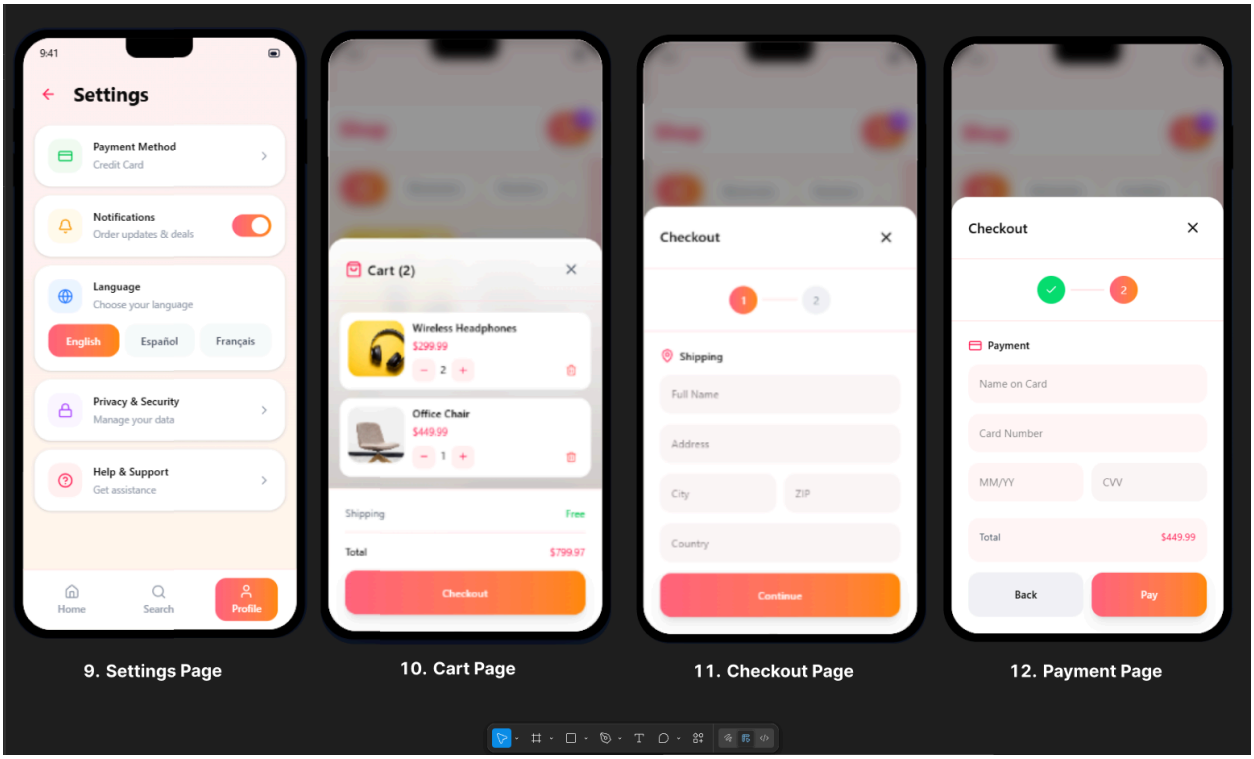


5. Search Page

6. Profile Page

7. My Orders Page

8. Wishlist Page

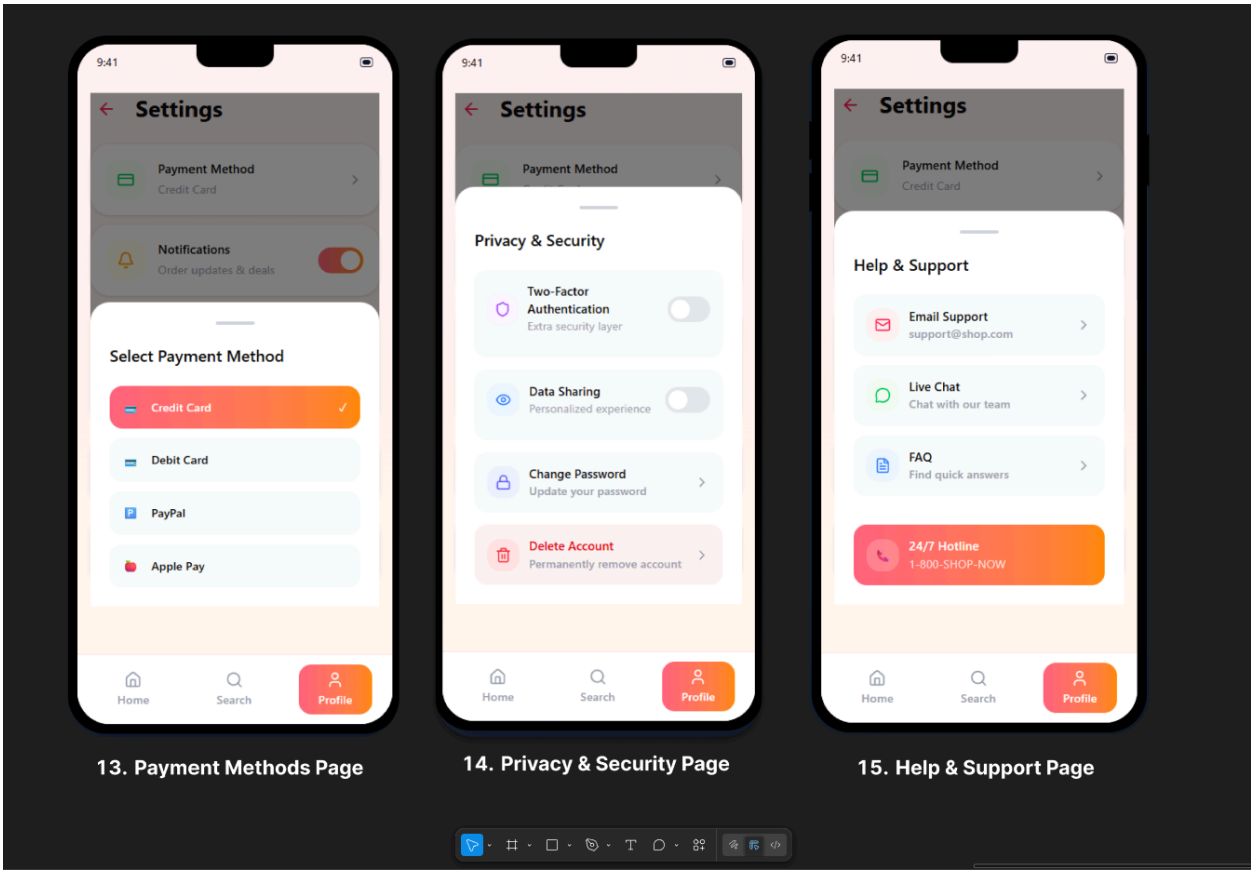


9. Settings Page

10. Cart Page

11. Checkout Page

12. Payment Page



7. Usability Considerations

7.1 Consistency

Consistent button styles, product cards, and iconography across screens.

7.2 Visibility of Important Elements

- Prominent “Add to Cart” button
- Large images
- Clear category labels

7.3 Feedback

- Highlighted button states
- Confirmation messages during checkout

- Error messages for form validation

7.4 Accessibility

Design included:

- High-contrast text
- Sufficient spacing for touch interactions
- Logical reading order
- Simple fonts for readability

7.5 Error Prevention

- Form validations (e.g., invalid card number warnings)
- Required-field indicators
- Cart quantity controls to avoid mistakes

8. User Evaluation & Insights

Five users explored the Figma prototype. Tasks included browsing, searching, viewing product details, and completing checkout.

Findings:

- **Navigation was smooth and intuitive** for most users.
- Users particularly liked the clean layout and minimalistic product cards.
- One user had difficulty locating the filter tool, prompting its repositioning.
- The checkout process was described as **“easy”** and **“straightforward.”**
- Breadcrumbs were added to improve orientation.

9. Conclusion

The final e-commerce app prototype demonstrates a strong application of HCI principles. The design was informed by user research, shaped with detailed personas, and improved through iterative testing. High-fidelity screens illustrate a polished, modern shopping experience emphasizing clarity, efficiency, and accessibility. The completed prototype provides a strong foundation for future development and further user testing.